Myke-Vital Temgoua Sao
Department of Physics, Faculty of Science
University of Yaoundé I
P.O. Box 812, Yaoundé, Cameroon
myke-vital.sao@facsciences-uy1.cm

August 28, 2026

Prof. Kenneth M. Merz, Jr., Editor-in-Chief
Journal of Chemical Information and Modeling
American Chemical Society

Dear Prof. Merz,

We submit our manuscript **"Resistance-Aware Docking Prioritization of Antimalarial Leads from African Natural Products"** for consideration as an Article in *JCIM*.

This work evaluates a docking-based resistance-aware prioritization strategy and uses targeted molecular dynamics as a separate structural follow-up; it does not assign a systems-level mechanism of action from docking alone. The 17 Set-C candidates were evaluated by docking-derived RRS across six resistance states, with ACSI and PNS providing complementary chemical-space and network-ranking dimensions; four separate parent-study complexes underwent targeted wild-type MD (10 ns each; 40 ns total). The target-balanced primary panel contained one Class A*, one Class A, four Class B, five Class C, and one Class D candidate; the five PfCRT-only records are reported separately as a coverage-limited sensitivity analysis. Only PfCRT–214 yielded an interpretable MM-GBSA estimate; the other parent-study systems were dissociated or conversion-excluded. Tartarus analysis provided no evidence of a composite MPO–affinity association, and this null result is interpreted as separation of ranking dimensions rather than proof of statistical independence.

A secondary 16-system Set-C MD pilot is reported only to quantify the extent to which trajectory-derived geometric and MM-GBSA summaries agree with docking-derived RRS. It is a single-replicate, 10-ns-per-system analysis of two candidates, is not merged into the primary docking-RRS estimand, and is not used as biological validation; mutant-to-wild-type ratios are interpreted strictly as retention within scoring noise rather than as gains. Lightweight post-selection diagnostics assessed candidate-removal and score-perturbation sensitivity, while companion P1 results are used only as upstream methodological and chemical-space context rather than independent replication.

Code, source data, and analysis records are available in the GitHub repository under an open-source licence. Statistical interpretation is bounded by effect sizes, seeded permutation p-values, bootstrap intervals, and the prespecified Bonferroni correction.

Suggested reviewers.

1. Dr. Fidele Ntie-Kang, University of Buea — African NP computational discovery
2. Dr. Kelly Chibale, University of Cape Town — antimalarial drug discovery
3. Dr. John D. Chodera, Memorial Sloan Kettering — free energy methods, MD best practices

This manuscript is original, not under consideration elsewhere, and all authors have approved the submission.

Sincerely,

Myke-Vital Temgoua Sao
(on behalf of all co-authors)